# Review: Identifying similar triangles

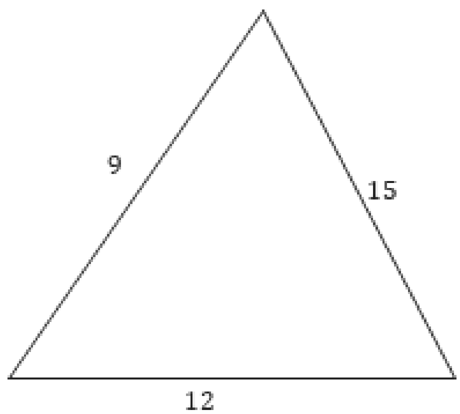
1 Consider the following sets of four triangles:

i Determine the similar triangles.

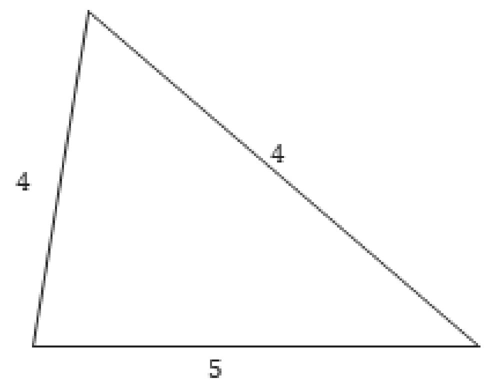
ii Give a suitable reason for their similarity.

a

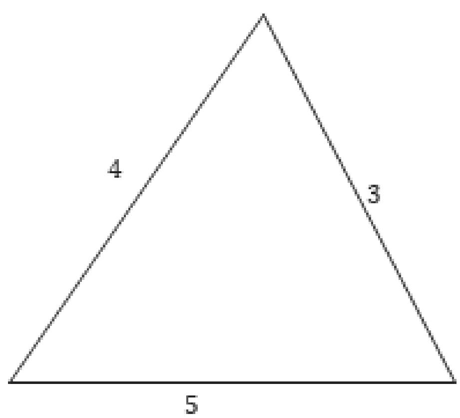
**A**



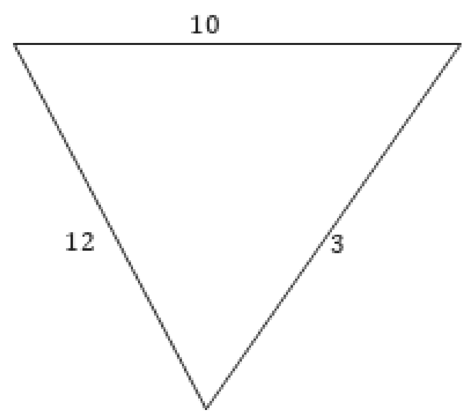

**B**




**C**

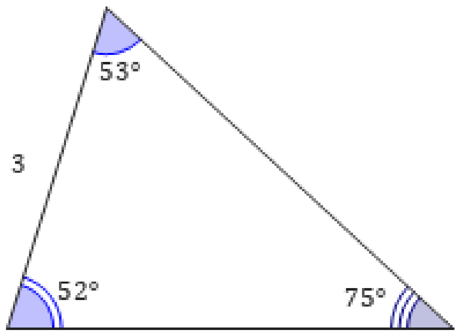



**D**

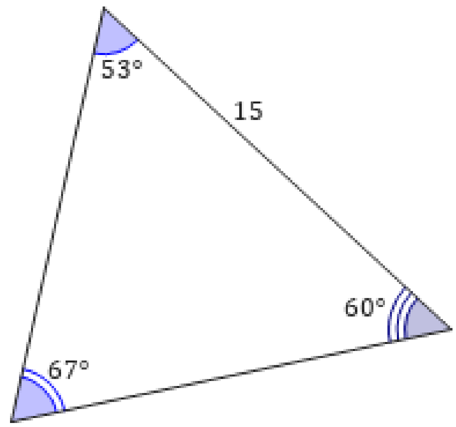


b

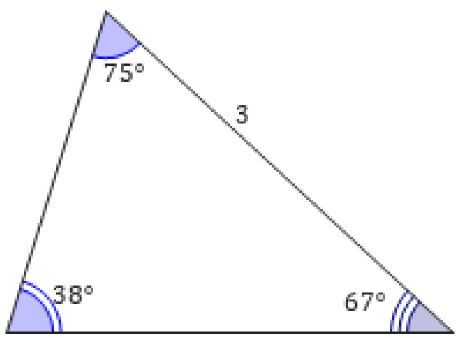
A



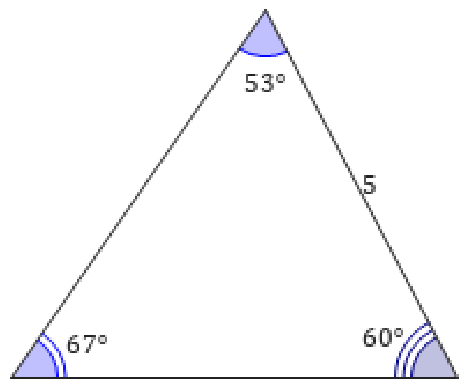

B




C

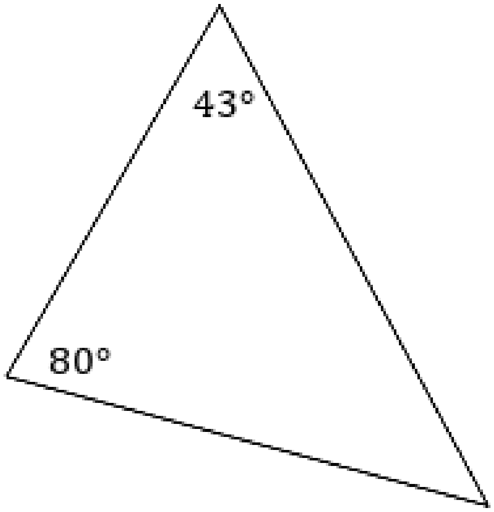



D

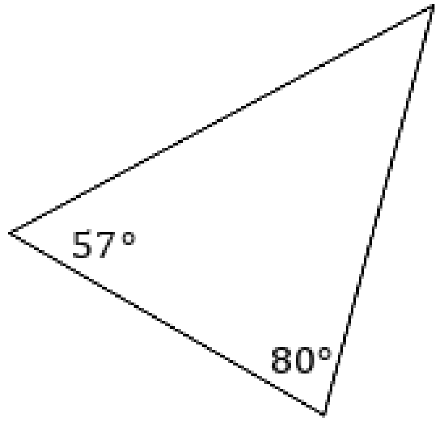


**2** Which of the following triangles are similar?

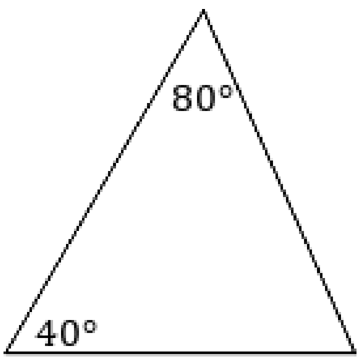
**A**



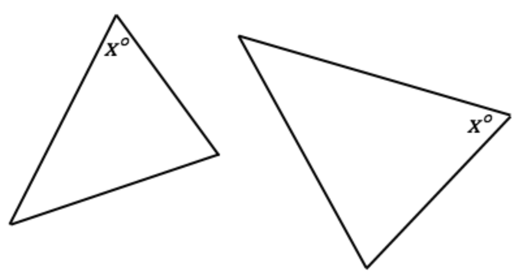

**B**



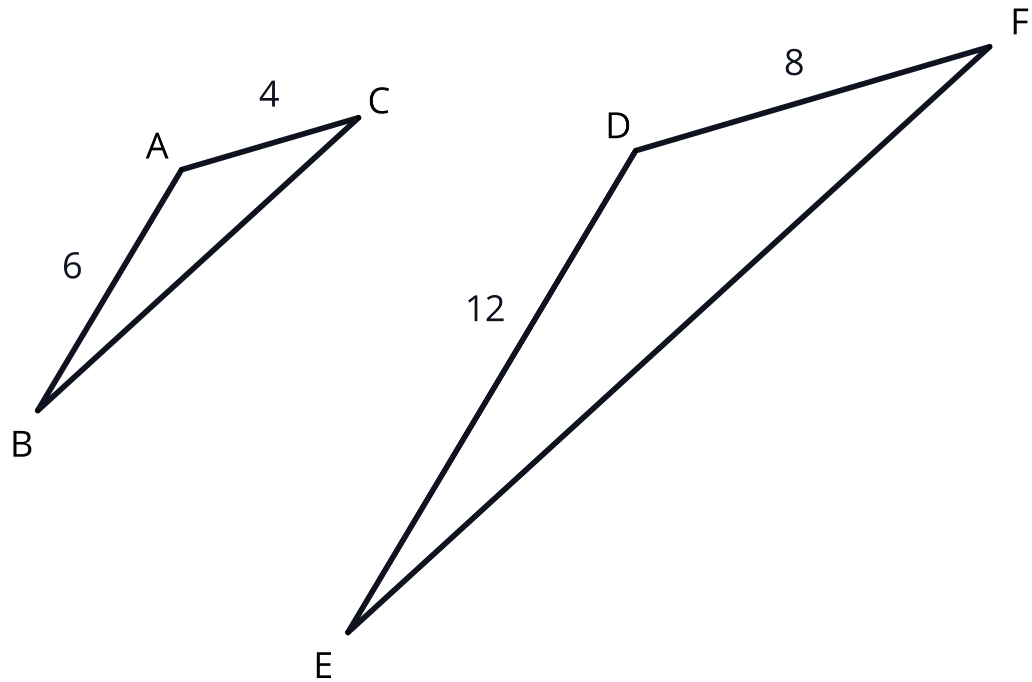

**C**




**3** Consider the following triangles.
What is the minimum amount of extra information we would need to establish that these are similar triangles.



**4** Consider the following triangles.




Determine whether the following information, if added to the diagram, could be used to prove immediately that the triangles are similar.

**a** $\angle ABC \cong \angle DEF$

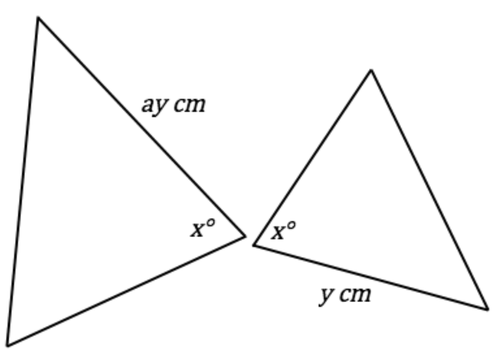
**b** $\angle BAC \cong \angle EDF$

**c** $\overline{BC} \cong \overline{EF}$

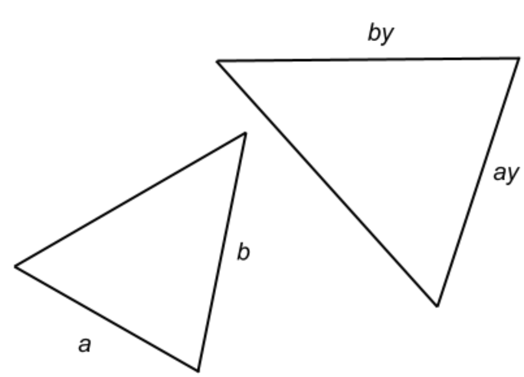
**d** $\angle ACB \cong \angle DFE$

**5** For each of teh given pairs of triangles, determine whether there are enough information to show whether ot not the triangles are similar.

**a**

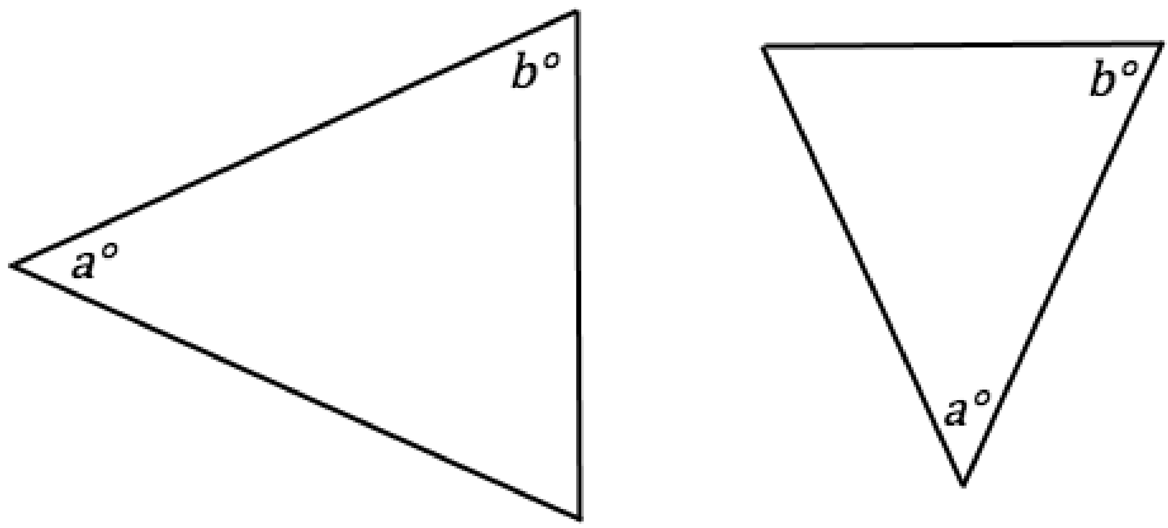



**b**

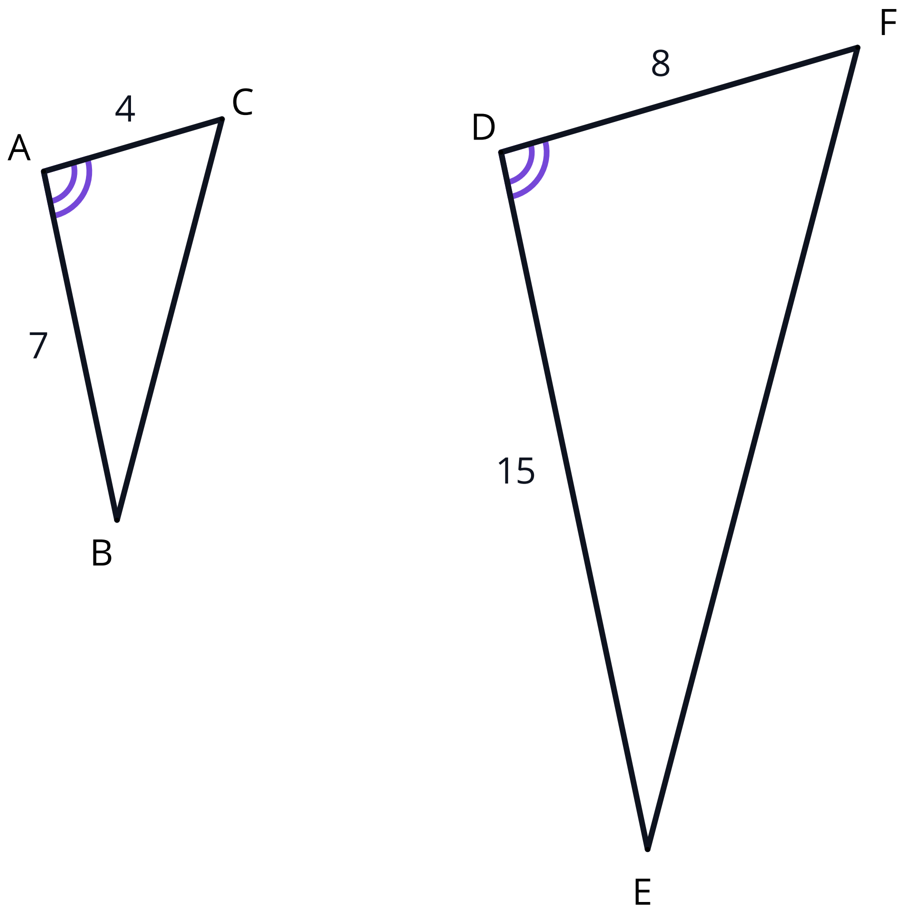


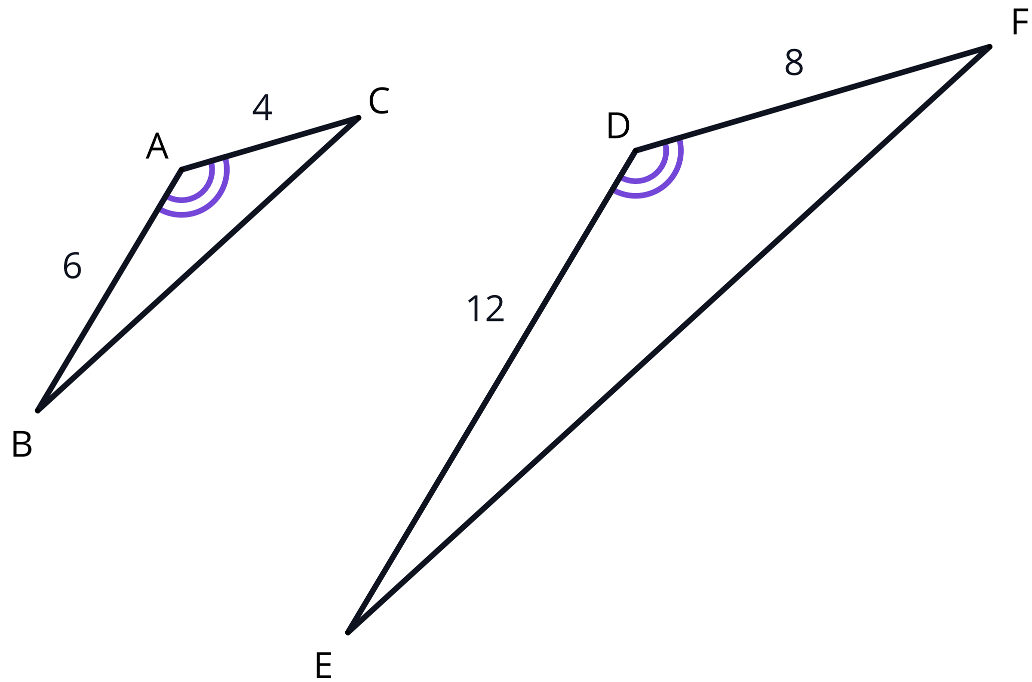
6 Determine whether the following pairs of triangles are similar or not. Give the reason for your answer.

a

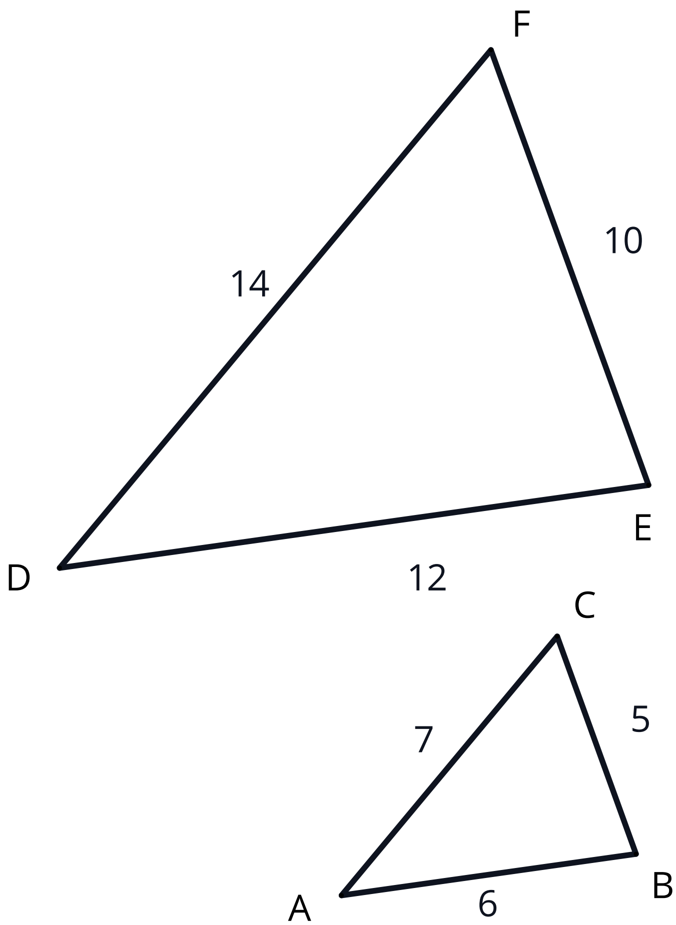


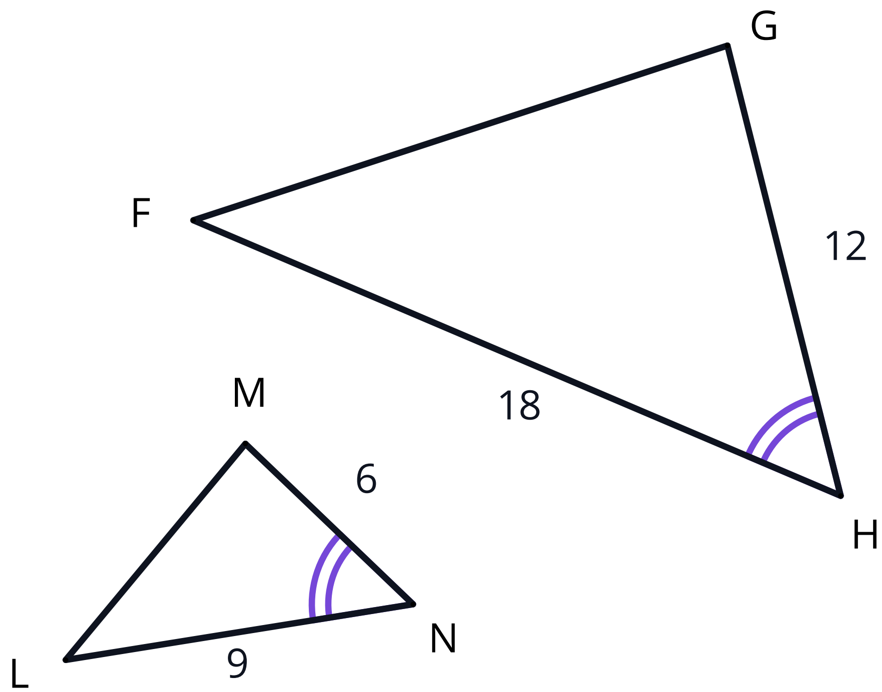
b

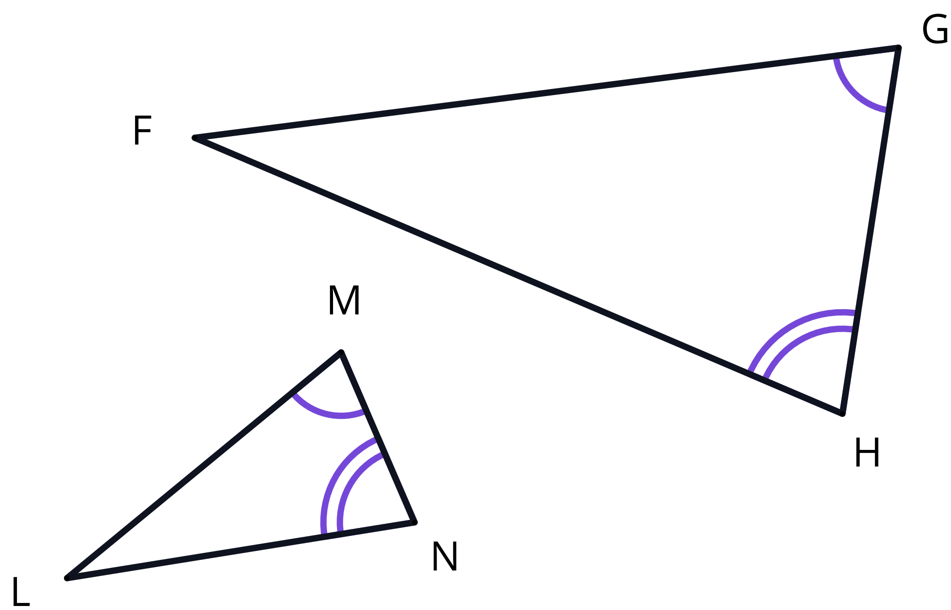


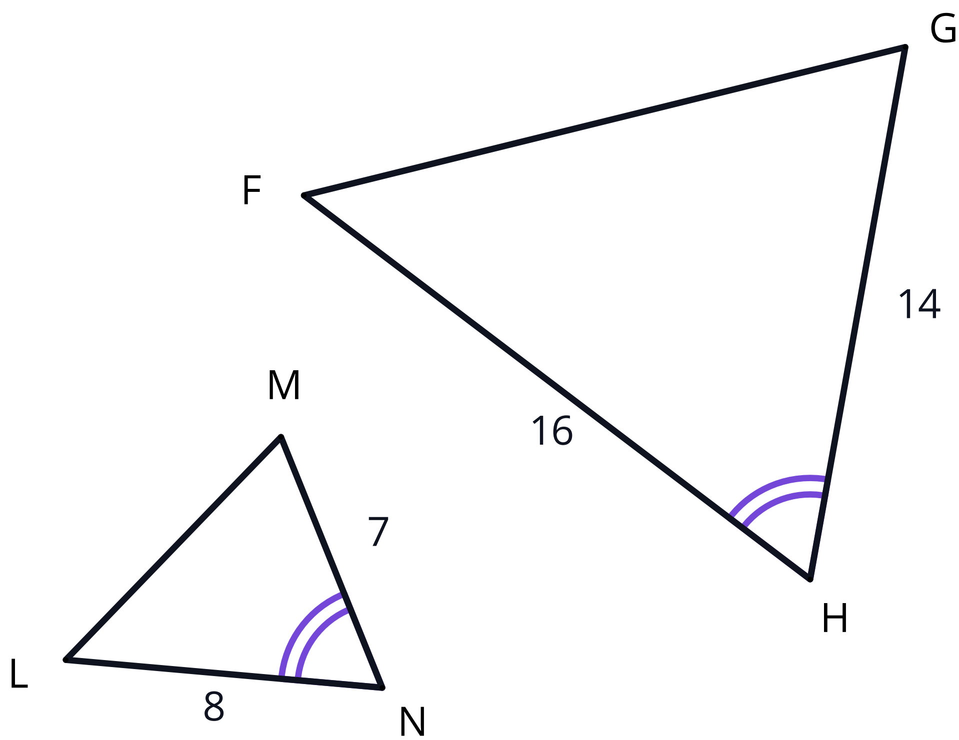

A
C
B
4
6
D
F
E
8
12

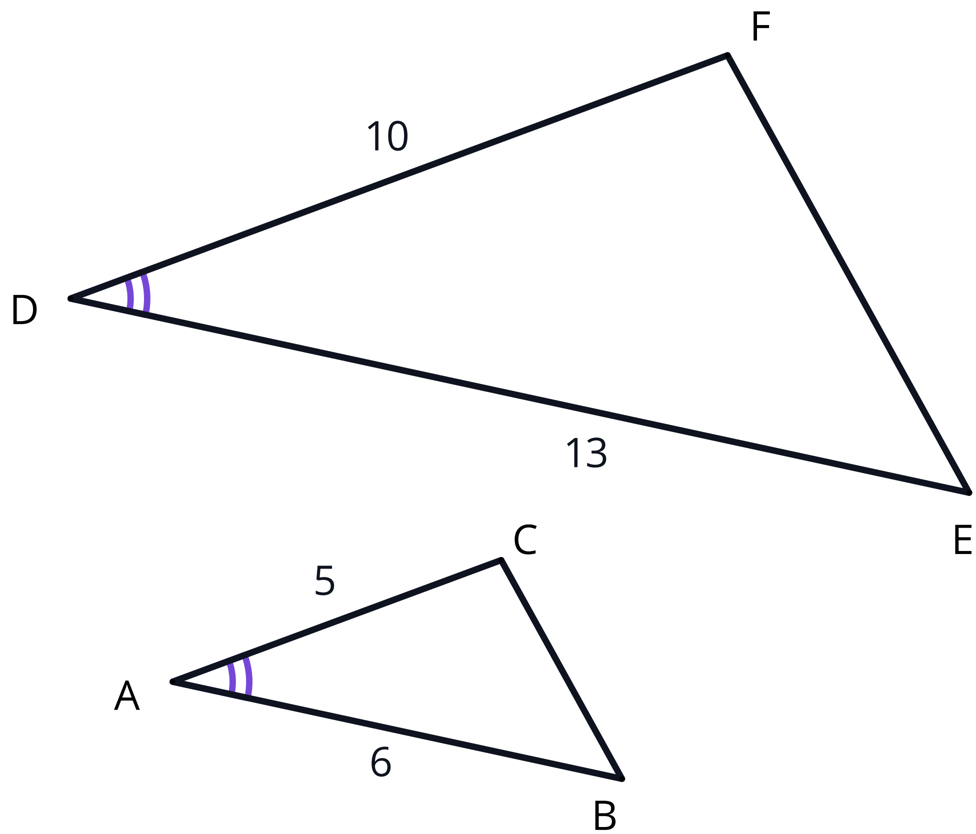
d




G
F
12
M
18
6
H
N
L
9


G
F
M
H
N
L


G
F
14
M
16
7
H
L
8
N



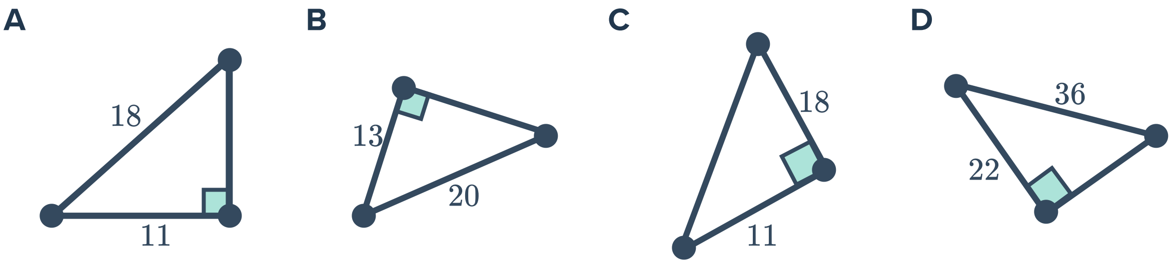

**7** Is it true that all equilateral triangles are similar? Justify your reasoning.

## Additional questions

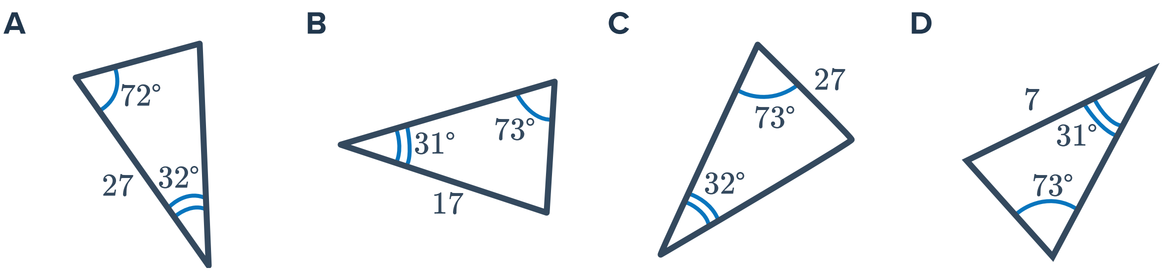
**8** For each of the following sets of triangles:

- **i** Identify which two are similar.
- **ii** Identify the similarity test that the pair satisfy: AA, SSS, SAS, or HL.

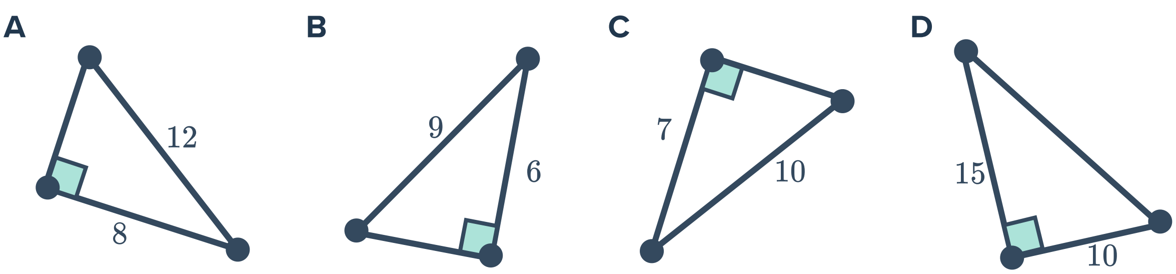
**a**



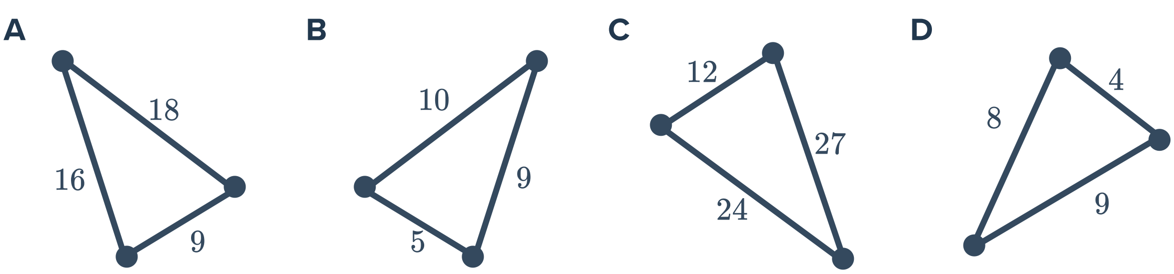

**b**




**c**

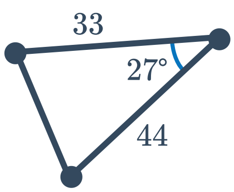



**d**

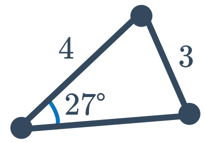


e

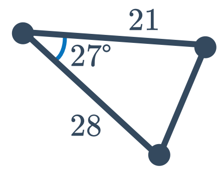
A



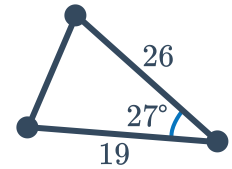

B




C

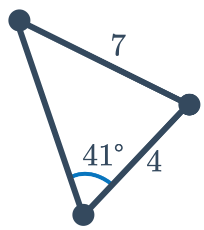



D

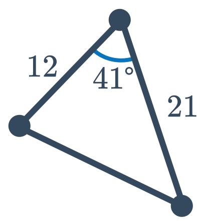



f

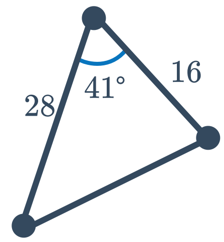
A



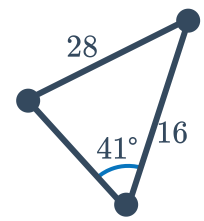

B




C

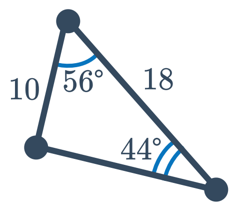



D

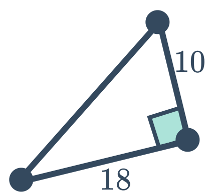



g

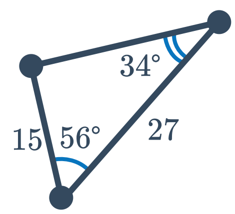
A



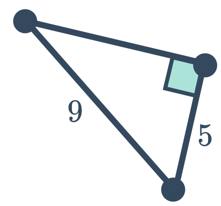

B




C

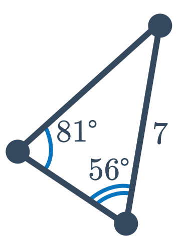



D

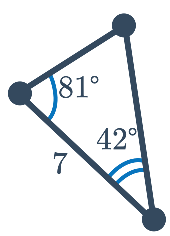



h

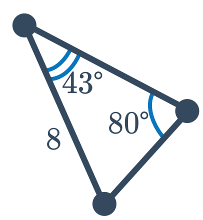
A



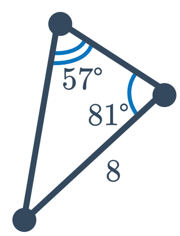

B




C

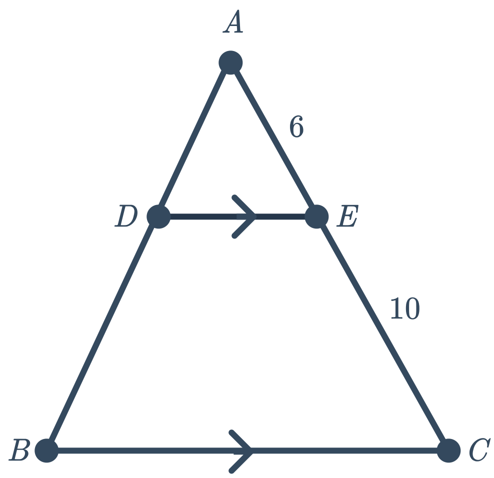



D

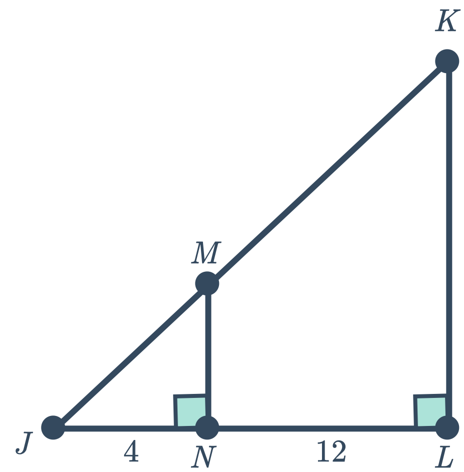


**9** For the following diagrams, identify the two [redacted]ar triangles and state the similarity test that they satisfy.

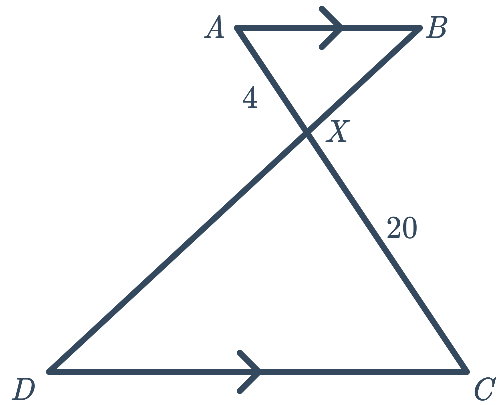
**a**



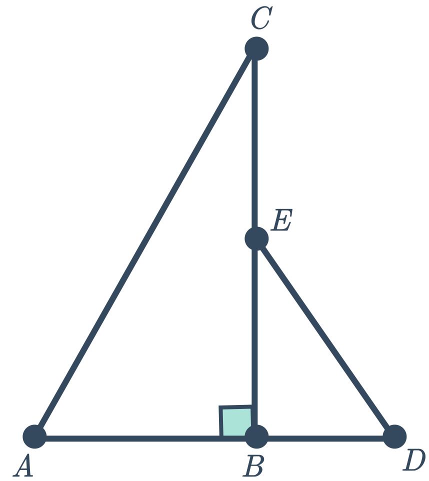

**b**



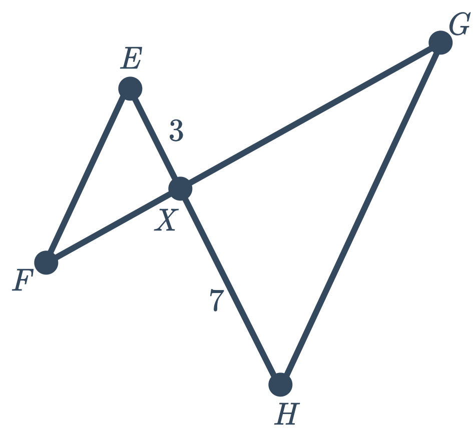

**10** Consider the following diagram:
State the similarity test that triangles $\triangle ABX$ and $\triangle CDX$ satisfy.



**11** Consider the following diagram, where $\overline{AD}$ is a straight line:
What information is required to know that triangles $\triangle ABC$ and $\triangle DBE$ are in fact similar?




**12** Consider the following diagram, where $\overline{EH}$ and $\overline{FG}$ are straight lines:
What information is required to know that triangles $\triangle EFX$ and $\triangle GHX$ are in fact similar?




**13** Consider the following diagram, where $\overline{TZ}$ and $\overline{UY}$ are non-parallel straight lines:

**a** Explain why the triangles $\triangle TUX$ and $\triangle YZX$ are not similar.

**b** What length does $YZ$ need to be for the two triangles to be similar?